

Challenges to be addressed

- Mushrooming of low-quality unauthorised production units producing sub-standard products and low cost of Chinese imports with poor quality result in low consumer confidence.
- Lack of awareness among consumers as well as institutional buyers involving government bodies about the efficiency of final products with respect to lux and watt, life expectancy, etc often result in the use of products with higher wattage but lesser efficiency.
- Due to the inability to produce LED chips and microchips, the country is still totally dependent on the import of chips at a high cost. This also puts constraints in developing a wider variety of LED lights with more colour options.
- Use of inefficient drivers results in enhanced energy usage as well as product failure.
- There is an acute shortage of experts from fields like chemical, electronics optics, lighting and thermal management, which are primarily required for manufacturing LED chips.

Further expansion of such initiatives across the country will open new market areas for the LED lighting industry. Retro-fit installations are in demand across all kinds of application areas.

Smart ways to grow

With rapid urbanisation and growing consumer awareness, the lighting industry foresees a growing demand for smart lighting solutions in Indian homes. Additionally, there is a growing potential for connected lighting solutions for offices and commercial buildings.

City streetlights have also become intelligent and can serve as a vital backbone for city administrators. An interactive city system allows the authorities to remotely monitor and control lights across the city from the control room itself. Faulty lights can be detected accurately through GPS technology in the pole and, hence, repairs can be carried out immediately, thereby reducing downtime. Lights can also monitor air quality, traffic, temperature and noise, and feed these to city administrators for analysis.

Lighting systems get smarter as a vision for autonomous, self-commissioning illumination systems is emerging. With the advent of LED lighting, the industry has transformed from analogue to digital, as LED lighting allows users to control, monitor and measure lighting output. This transformation is taking place across public, home and professional lighting, and smart connected LED lights will emerge as the largest Internet of Things (IoT) device in the next five to

ten years. Control devices, dimmers and wireless lighting with advanced sensors will cater to the needs of modern consumers.

According to a study by Nielsen, lighting control is the most frequently used daily home automation feature. Smart lights appear on a shortlist of technologies beginning to appear on many buyers' must-have lists because of their ability to improve security and comfort, and cut electricity costs. According to Gartner, smart lighting applications are expected to continue growing to reach 2.54 billion units installed by 2020 globally.

Smart lighting can be considered more like a service than a product, as it ensures a better experience. For example, instead of switches and push buttons, motion sensors are used to control lights automatically. This, in turn, reduces human intervention and saves energy. Smart lighting also offers more convenience in other ways, such as a change in colour as per weather conditions, light pulse to replace the doorbell sound and so on.

Leading brands are paving the way with smart LED bulbs and switches that connect to Wi-Fi and offer app, button and voice controls over positioning, brightness, warmth and colour of connected bulbs. The major technology behind smart connected homes is the IoT and, according to Strategy Analytics, IoT growth is higher than ever, but annual growth will begin to tail off as the market normalises in the near future, dropping to nine per cent by 2021.

In a study, it was found that most

of the consumers are worried about the security of their home while they are away. Smart homes have a solution for this, too. Lights that come on at certain times of the day—indoors and outdoors—give the illusion that occupants are at home and create a more secure environment. There are multiple ways to customise this, too. As a homeowner drives up to his or her home, lights can come on with geofencing. Alternately, he or she can remotely use an app to turn the lights on. It can be as simple having lights on a timer, too.

Currently, smart lighting technology has been adapting to seamlessly integrate into smart home systems. Bluetooth-enabled systems that do not rely on the Internet need only the app and the fixture to operate. Even light switches and bulbs may look completely different in ten to fifteen years as consumers are looking for cleaner and sleeker designs.

Moving forward

The digital nature of LED technology has brought illumination and IT together, allowing lighting systems to participate in the IoT. This has further led to the emergence of connected lighting, marking a significant shift and transforming lighting from a commodity product to a fully-integrated lighting system that can seamlessly connect with a wireless network or Ethernet, allowing users to remotely control and monitor their lighting systems. Thus, new LED lighting systems will be an outcome of right partnerships between traditional players and IT enablers.

Introduction of Light-Fidelity (Li-Fi) technology will also help the industry evolve to a great extent. This high-quality LED lighting-based technology provides a broadband Internet connection through light waves, thereby reducing exposure to an electromagnetic environment, currently caused by radio waves and Wi-Fi.

However, rather than following what developed countries have done in smart or connected lighting, the Indian lighting industry needs to focus on ease-of-use and price-sensitivity issues for the Indian market. **EFY**